

FortiGate Syslog Integration and Incident Correlation in Wazuh

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1. Purpose

This SOP describes the step-by-step procedure to integrate FortiGate firewall logs into Wazuh, validate log ingestion, create custom detection rules, and implement incident correlation for SOC monitoring.

2. Scope

This procedure applies to:

- Wazuh Manager deployments
 - FortiGate firewall syslog forwarding
 - SOC monitoring and threat detection use cases
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3. Prerequisites

3.1 Infrastructure Requirements

- Wazuh Manager installed and running
- FortiGate firewall with syslog capability
- Network connectivity between FortiGate and Wazuh server
- UDP port 514 reachable

3.2 Access Requirements

- Root or sudo access on Wazuh server
 - Admin access to FortiGate
 - Wazuh Dashboard access
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4. Architecture Overview

FortiGate → Syslog (UDP 514) → Wazuh Manager → Decoder → Rules → Alerts → Dashboard

5. Configuration Steps

5.1 Configure FortiGate Syslog

On FortiGate CLI:

```
config log syslogd setting
  set status enable
  set server <WAZUH_IP>
  set mode udp
  set port 514
end
```

Verify logs are enabled for:

- Traffic
 - Event
 - System
 - VPN
-

5.2 Configure Wazuh to Receive Syslog

Edit:

/var/ossec/etc/ossec.conf

Add or verify:

```
<remote>
  <connection>syslog</connection>
  <port>514</port>
  <protocol>udp</protocol>
  <allowed-ips>192.168.0.1</allowed-ips>
</remote>
```

Restart Wazuh:

```
sudo systemctl restart wazuh-manager
```

6. Validation Procedures

6.1 Verify Port Listening

```
sudo ss -ltnp | grep 514
```

Expected: UDP 514 listening.

6.2 Verify Network Traffic

```
sudo tcpdump -i any port 514
```

Expected: FortiGate packets arriving.

6.3 Verify Log Ingestion

```
sudo tail -f /var/ossec/logs/archives/archives.log
```

Expected: Raw FortiGate logs visible.

6.4 Verify Alert Generation

```
sudo tail -f /var/ossec/logs/alerts/alerts.log
```

Expected: Parsed Forti alerts.

7. Custom FortiGate Detection Rules

Create file:

```
/var/ossec/etc/rules/fortigate_custom_rules.xml
```

7.1 Base Detection Rule

```
<group name="fortigate,custom,">
```

```
<rule id="110600" level="10">
```

```
<if_sid>81619</if_sid>
```

```
<description>FortiGate suspicious traffic detected</description>
```

```
<mitre>
```

```
<id>T1110</id>
```

```
</mitre>
```

```
</rule>
```

```
<rule id="110601" level="12">
```

```
<if_group>fortigate</if_group>
```

```
<field name="type">event</field>
```

```
<field name="subtype">system</field>
```

```
<description>FortiGate admin/system activity detected</description>
</rule>

<rule id="110602" level="14" timeframe="300" frequency="2">
  <if_matched_sid>110600</if_matched_sid>
  <if_matched_sid>110601</if_matched_sid>
  <description>CRITICAL: Suspicious traffic followed by admin activity</description>
  <group>incident,fortigate,critical,</group>
  <mitre>
    <id>T1068</id>
  </mitre>
</rule>

</group>
```

8. Rule Validation

Run:

```
sudo /var/ossec/bin/wazuh-analysisd -t
```

Expected output:

Configuration OK

Restart manager:

```
sudo systemctl restart wazuh-manager
```

9. Dashboard Verification

Navigate to:

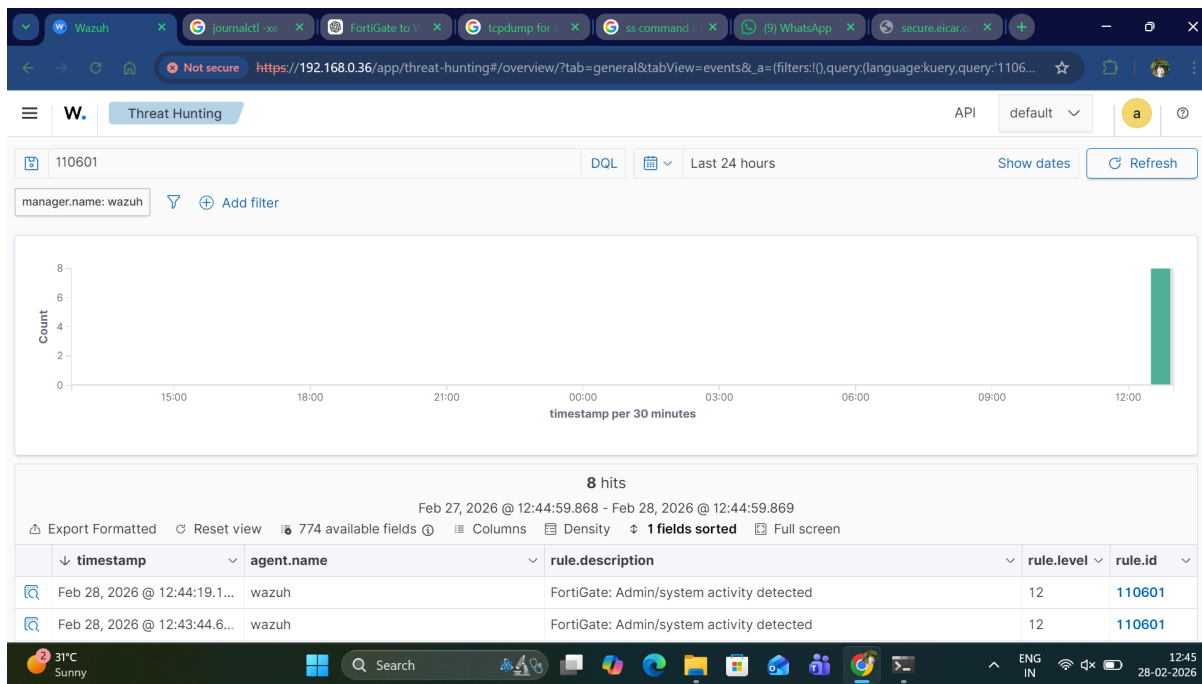
Wazuh Dashboard → Threat Hunting

Test filters:

rule.id:110601

rule.id:110602

rule.groups: fortigate



Expected:

- Admin/system alerts visible
- Correlated incidents when conditions met

10. Incident Testing Procedure

10.1 Generate Traffic Burst

From a client machine:

for i in {1..30}; do curl https://google.com; done

10.2 Generate Admin Activity

- Login to FortiGate GUI
- Logout
- Make minor config change

10.3 Expected Result

Wazuh should trigger:

- Rule 110600
- Rule 110601
- Rule 110602 (CRITICAL)

11. Troubleshooting Guide

Issue: Logs seen in tcpdump but not in Wazuh

Check:

- allowed-ips
- port 514 listening
- firewall rules

Issue: Decoder not matching

Check:

- Forti log format
- Wazuh built-in Forti decoders

Issue: Manager fails to start

Check:

- Duplicate rule IDs
- XML syntax
- Missing group wrapper

Issue: Correlation not firing

Check:

- frequency/timeframe
 - both prerequisite rules firing
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12. Security Considerations

- Restrict allowed-ips to firewall IP only
 - Use high custom rule ID ranges (110000+)
 - Avoid modifying default Wazuh rules
 - Validate rules before restart
-

13. Maintenance

Daily

- Monitor Security Events
- Review critical alerts

Weekly

- Tune false positives

- Review Forti log volume

Monthly

- Update MITRE mappings
 - Review rule effectiveness
-

14. Conclusion

This SOP enables:

- Centralized FortiGate monitoring
- Automated threat detection
- SOC-level incident correlation
- Improved security visibility

The implementation is production-ready when all validation steps pass and correlated alerts are observed in the Wazuh dashboard.

End of Document